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Autonomous Border Surveillance System

An Interdisciplinary Project Report (AI367P)

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RV College of Engineering®

**In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments**

2025-26



RV College of Engineering®, Bengaluru
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CERTIFICATE

Certified that the interdisciplinary project (AI367P) work titled ***Autonomous Border Surveillance System*** is carried out by _____ of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

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DECLARATION

We, **N Mohammed Akhil, B Hanumanth, K V Sai Charan, Tanmay Sharad Joshi** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Autonomous Border Surveillance System**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

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Our sincere thanks to the project coordinator \projectCoOrdNameA currently not set. Please fix this., Professor, Department of \projectCoOrdDeptAShortForm in not defined. Fix this by placing inside square brackets. for timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

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ABSTRACT

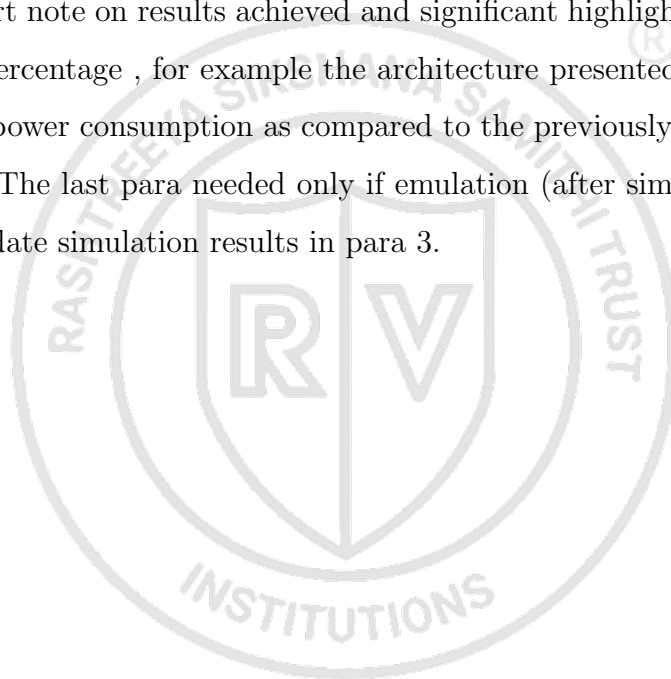
Highlights of significant contributions: One page with 3 to 4 paragraphs

Paragraph 1: Importance and relevance of Topic, reported issues and limitations of the topic in performance or computation etc, issues involved in those limitations, need for addressing those issues and a short note on how that is addressed in this report.

Paragraph 2: Objectives of this work, short note on algebraic methods used and formulations achieved, computational procedures developed. Integrated Circuit (IC).

Paragraph 3: Description of simulation procedure including SW tools used and choice of test cases. Short note on results achieved and significant highlights of improvements if any in terms of percentage , for example the architecture presented in this report shows 22 % improve in power consumption as compared to the previously reported articles. IC

Paragraph 4: The last para needed only if emulation (after simulation) or Hardware developed to validate simulation results in para 3.



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Chapter 1

Autonomous Border Surveillance

System

CHAPTER 1

AUTONOMOUS BORDER SURVEILLANCE SYSTEM

Autonomous surveillance systems are becoming essential in modern security infrastructures where continuous monitoring, rapid response, and minimal human intervention are required. This chapter introduces the concept, motivation, and objectives of the proposed Autonomous Border Surveillance System. It also discusses the problem statement, methodology, assumptions, and overall organization of the report.

1.1 Introduction

Border security has traditionally relied on human personnel, CCTV monitoring systems, and manual patrols. These approaches are often limited by human fatigue, delayed response times, and the inability to continuously monitor large areas efficiently. With advancements in embedded systems, wireless communication, and sensor technologies, it is now possible to design intelligent systems capable of detecting, identifying, and responding to intrusions autonomously.

The proposed project, titled **Autonomous Border Surveillance System**, aims to address these limitations by developing a smart embedded system capable of real-time intrusion detection and response. The system utilizes ultrasonic sensing for detection, servo motors for directional scanning and targeting, RFID technology for identification of authorized personnel, and LoRa communication for long-range alert transmission. The system is further designed to integrate advanced computer vision techniques in future phases for enhanced accuracy.

This project combines concepts from embedded systems, communication engineering, and intelligent automation to create a prototype surveillance solution suitable for border monitoring and restricted area protection.

1.2 Literature Review

Autonomous surveillance systems have been widely studied in recent years, particularly in the context of defense and security applications. Several research works have focused on sensor-based detection mechanisms using ultrasonic, infrared, and microwave sensors. These systems provide real-time detection but often suffer from false positives

due to environmental noise.

Wireless communication technologies such as LoRa have been explored extensively for long-range, low-power communication in surveillance applications. Studies indicate that LoRa provides reliable communication over several kilometers with minimal power consumption, making it ideal for remote monitoring systems.

RFID-based identification systems have been implemented in access control and security systems to differentiate between authorized and unauthorized individuals. These systems provide fast and reliable identification but require integration with detection systems for real-time decision-making.

Recent advancements in computer vision, particularly deep learning models such as YOLO (You Only Look Once), have significantly improved object detection and classification accuracy. These models are capable of identifying humans and objects in real time, making them suitable for integration into surveillance systems.

Despite these advancements, there remains a gap in fully integrated systems that combine detection, identification, targeting, and communication into a single autonomous platform. The proposed system aims to bridge this gap by integrating multiple technologies into a cohesive and efficient solution.

1.3 Motivation

The motivation for this project arises from the increasing need for efficient and reliable border surveillance systems. Traditional methods are labor-intensive and prone to human error, especially in harsh environments where continuous monitoring is required.

The growing availability of low-cost microcontrollers, sensors, and communication modules makes it feasible to design intelligent systems that can operate autonomously. Additionally, the need for rapid threat detection and response in sensitive areas such as borders, military zones, and restricted facilities further emphasizes the importance of such systems.

This project aims to explore the practical implementation of an autonomous surveillance system that can detect intrusions, classify targets, and communicate alerts in real time, thereby enhancing security and reducing dependency on manual monitoring.

1.4 Problem Statement

Existing border surveillance systems rely heavily on manual monitoring and isolated technologies, leading to delayed response times and increased chances of security breaches. There is a need for an integrated system that can autonomously detect intrusions, identify authorized personnel, and communicate alerts efficiently over long distances. The challenge lies in designing a system that is accurate, reliable, and capable of operating in real-time with minimal human intervention.

1.5 Objectives

The objectives of the project are:

1. To design and develop an autonomous surveillance system capable of detecting intrusions using ultrasonic sensing.
2. To implement identification mechanisms using RFID technology to differentiate between authorized and unauthorized entities.
3. To establish reliable long-range communication using LoRa technology for real-time alert transmission.

1.6 Brief Methodology of the Project

The proposed system follows a structured methodology involving detection, identification, targeting, and communication. Initially, the ultrasonic sensor scans the environment using a servo motor to detect objects within a predefined range. Once a target is detected, geometric calculations are used to determine its position and align the system accordingly.

Following detection, the system checks for RFID tags to identify whether the target is authorized. If the target is unauthorized, the system triggers an alert mechanism and transmits the information using LoRa communication to a remote monitoring station. The methodology ensures real-time operation and efficient decision-making.

The overall system flow can be summarized as follows:

- Continuous scanning using ultrasonic sensor
- Detection of object within threshold range

- Position calculation and targeting
- RFID-based identification
- Alert generation and LoRa transmission

1.7 Assumptions made / Constraints of the project

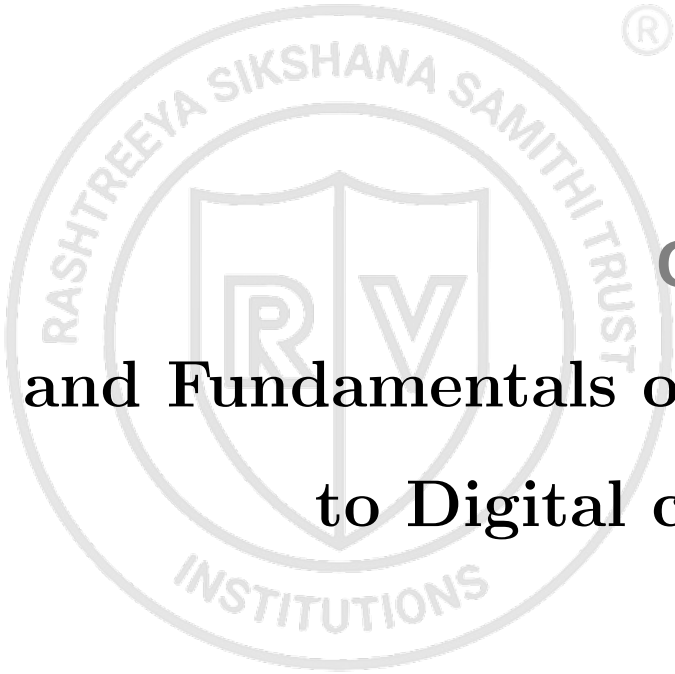
The following assumptions and constraints are considered in this project:

- The detection range is limited to the effective range of the ultrasonic sensor (approximately 30 cm in current implementation).
- RFID identification requires the authorized user to be within close proximity of the reader.
- Environmental factors such as temperature, humidity, and obstacles may affect sensor accuracy.
- The system operates under controlled conditions and may require calibration for real-world deployment.
- Power supply stability is assumed for continuous operation.

1.8 Organization of the Report

This report is organized as follows:

- Chapter 2 discusses the system design, architecture, and hardware components used in the project.
- Chapter 3 explains the methodology and implementation details of the system.
- Chapter 4 presents the results, observations, and performance analysis.
- Chapter 5 discusses challenges faced and improvements made during development.
- Chapter 6 concludes the report and outlines future enhancements.



Chapter 2

Theory and Fundamentals of Analog to Digital converter

CHAPTER 2

THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}  
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

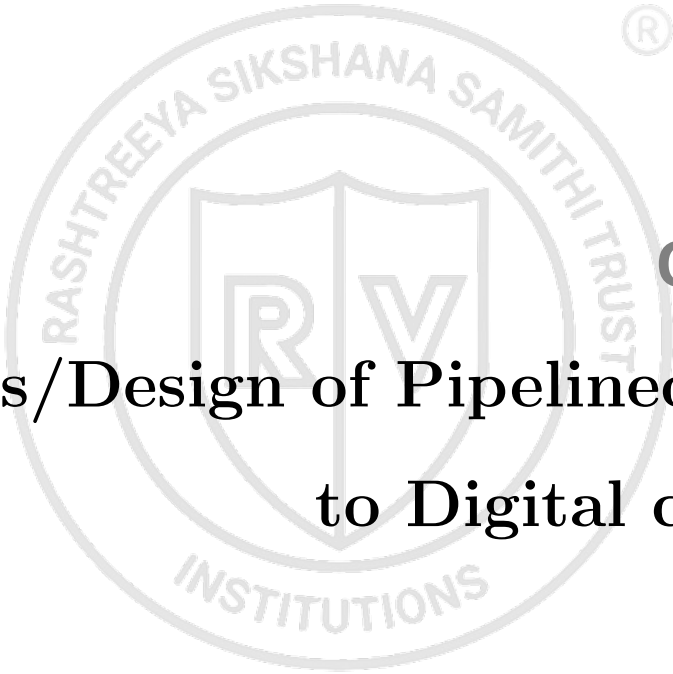
Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}  
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

The logo is a circular emblem. The outer ring contains the text "RASHTREEYA SIKSHANA SAMITHI TRUST" at the top and "INSTITUTIONS" at the bottom. In the center is a shield divided vertically, with a large "R" on the left and a large "V" on the right. A registered trademark symbol (®) is located to the upper right of the logo.

Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

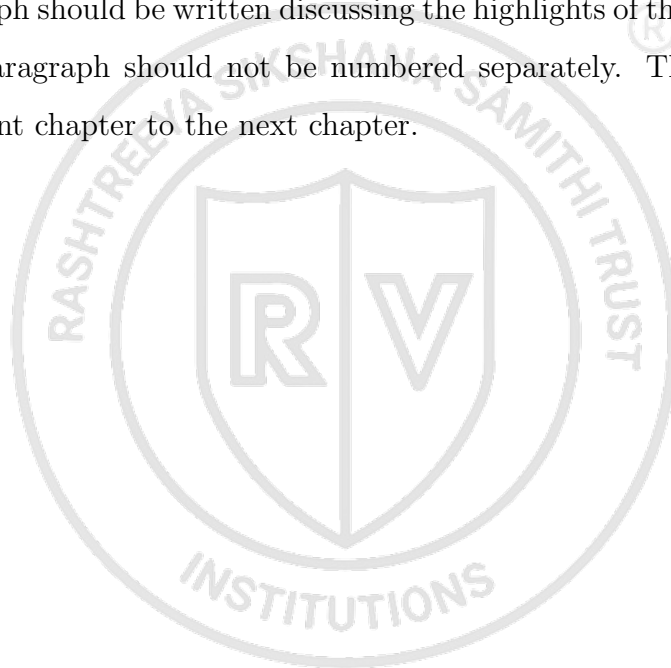
Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

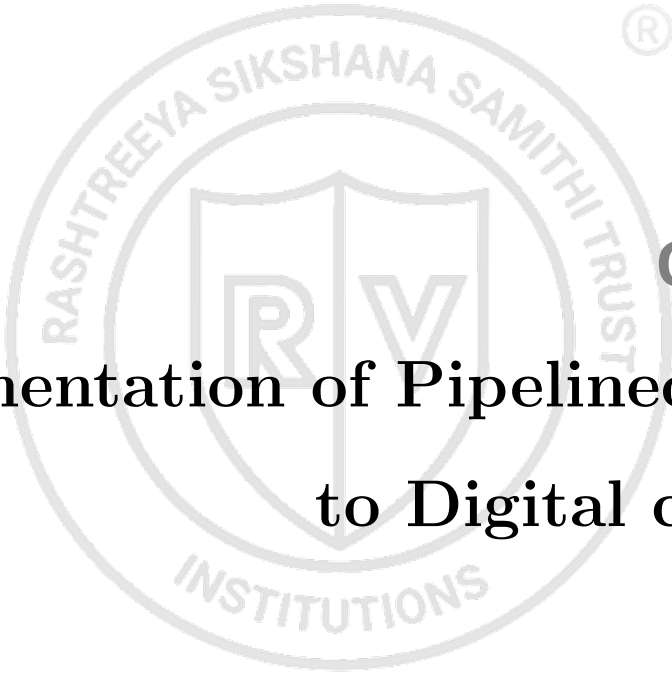
3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

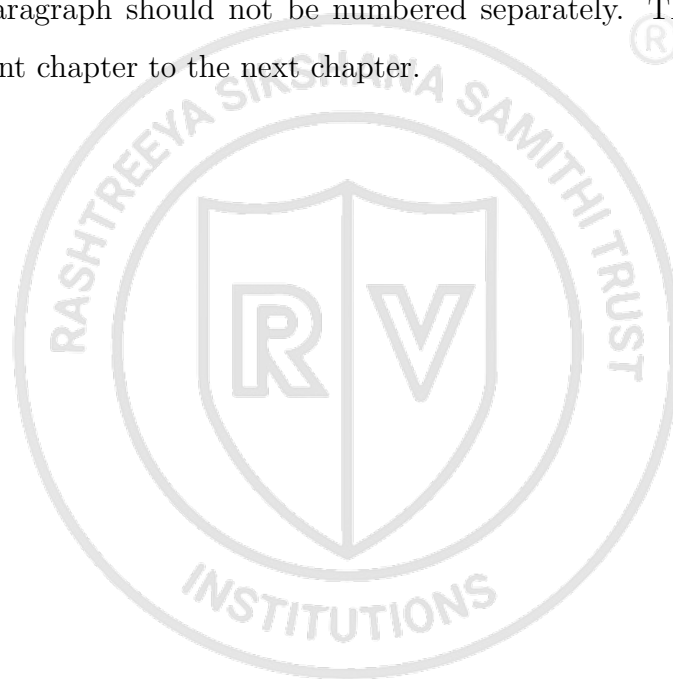
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```